

PARTS LIST

Semiconductors:

- IC1 - LM358 op-amp
IC2 - ATmega16 MCU

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 10-kilo-ohm
R5, R7-R12, R14 - 4.7-kilo-ohm
R6 - 1-kilo-ohm
R13 - 560-ohm
VR1 - 10-kilo-ohm preset
VR2, VR3 - 5-kilo-ohm potmeter
VR4 - 5-kilo-ohm preset
VR5 - 100-kilo-ohm potmeter

Capacitors:

- C1-C5, C7 - 0.1 μ F ceramic disk
C6 - 10 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector for power supply
CON2 - 2-pin female Berg connector for input signal
CON3 - 6-pin female Berg connector for ISP port
- 20-pin female Berg connector for GLCD
S1-S2 - On/off switch
S3 - Tactile switch
LCD1 - 128 $\times$ 64 GLCD
XTAL1 - 16MHz crystal oscillator
- 5V DC power supply

Adjust VR2 and VR3 to obtain about 50Hz frequency at the bottom, on the left side of the screen. One cycle

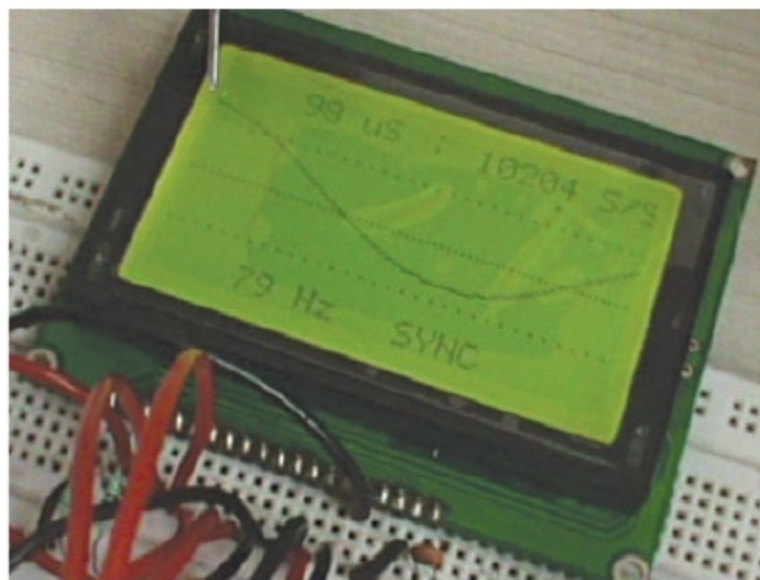


Fig. 5: Waveform display in GLCD

of the AC waveform will be seen on the screen. To have static waveform, press S2. It will start displaying waveform from maximum voltage level. To stop reading input samples, close S1. You can press reset switch S3 momentarily at any point of time if there is a problem in the display.

Note. 1. GLCD scope can read 14285 (maximum) to 28 (minimum) samples per second. The value may have a slight variation due to the time

taken to read ADC value by the MCU.

2. Sampling time cycle is available at pin 22 and total sampling (256) time cycle is available at pin 23 of the MCU. You can compare these signals using a professional oscilloscope.

3. It takes more time to read the signal as samples per second go down, that is, sample delays are increasing.

4. Use 5V rechargeable battery (power bank) as power supply to make the system handy.

5. Use 10-turn potmeter (preferred) as VR2 for easy control. **EFY**



Fayaz Hassan is manager at Visakhapatnam Steel Plant, Andhra Pradesh. He is interested in MCU projects, mechatronics and robotics



One company. Many solutions.



PREMIUM 3D PRINTER FILAMENTS

Small spool sizes so you can experiment with more colors without spending a bundle!

- 0.25kg spools
- 0.50kg spools



ENCAPSULATING AND POTTING EPOXIES

SILVER CONDUCTIVE EPOXY

Cold Solder Repair

Cat. No. B331

Now available in 40 gram kits!



PENETRATING OIL

Cat. No. 8472



TOTAL GROUND

Carbon Conductive Coating

Cat. No. 838

Create your own static-free work station



ACRYLIC CONFORMAL COATING

Cat. No. 419C



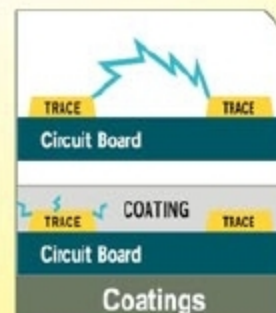
Flux Removers



Contact Cleaners



Circuitboard Protection



Coatings



EMI/RFI Shielding



No Clean Flux



Distributed in India by:

PROGRESSIVE ENGINEERS

51-A, DOLI CHAMBERS, STRAND ROAD, COLABA, MUMBAI - 400 005. PH.: 022-25006762 MOB.: +91-8898340064
VAPI OFF.: 0260-2451744 EMAIL: info@progressiveengineers.net WEBSITE: www.progressiveengineers.net